ELEC 231 — Mathematical Methods I for ECE · Fall 2026

Pre-tutorial 5

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This tutorial covers selected content from Chapter 4 of Murray R. Spiegel's *Applied Differential Equations*.

Problem

Problem 1 (Method of undetermined coefficients). Use the method of undetermined coefficients to solve the second-order ordinary differential equation

$$y'' + y = 6\cos^2(t), \tag{T5.1}$$

given

$$y(0) = 0, \qquad y(\pi/2) = 0. \tag{T5.2}$$

Assume that
(i) the general solution to the equation

$$y'' + y = 0, \tag{T5.3}$$

is

$$y_c = c_1 \sin(t) + c_2 \cos(t), \tag{T5.4}$$

and
(ii) a particular solution of (T5.1) is

$$y_p(t) = A + B\cos(2t) + D\sin(2t) \tag{T5.5}$$

where the coefficients A, B, D are to be determined. Solve (T5.1) for $y(t)$, given the conditions (T5.2).